STA 240: Probability and Statistics

Spring 2026 · Duke University

Course Overview

Description

This is a course on the mathematics of probability and statistics. In **probability**, we describe the distribution of a random phenomenon and then study how the realizations of that phenomenon typically behave. In **statistics**, we do the reverse; we observe realizations of a random phenomenon with *unknown* distribution, and then use the data to figure out what the distribution is. We will spend twelve weeks on the first, and then three weeks on the second. Topics include set theory, probability spaces, counting methods, conditional probability, discrete and (absolutely) continuous random variables, transformations of random variables, (pseudo)random number generation, bivariate distributions, concentration inequalities, limit theorems, maximum likelihood estimation, and Bayesian inference with conjugate priors.

Aside from the concrete topics, the course emphasizes four generic intellectual themes:

- Students in this class will improve their mathematical maturity. They will deepen their experience with all of the main ideas and techniques of univariate calculus, encounter topics like set theory and combinatorics, and get a taste of rigorous mathematical proof.
- Probability and statistics weasel their way into pretty much everything these days, and students will study a wide variety of famous and obscure applications coming from the natural and social sciences, technology and industry, and even arts and culture.
- Probability and statistics are often counterintuitive, and humans frequently make silly and harmful mistakes in reasoning. Students will use their mathematical knowledge to identify and critique these errors—and hopefully avoid them themselves.
- This is a course about doing the math, but in the modern era, this has a symbiotic relationship with computer simulation. Students will learn the basics of the R programming language and use it to simulate probabilistic environments and reinforce their mathematical reasoning.

Prerequisites: single-variable calculus.

Meetings

Meeting	Location	Time	Staff
Lectures	Old Chem 116	Mo/We 10:05 – 11:20 AM	Anya

Lab 01	Old Chem 201	Th 1:25 – 2:40 PM	Gwen
Lab 02	Social Sciences 105	Th 3:05 – 4:20 PM	Federico

Teaching Team

Name	Role	Office Hours
Anya Katsevich	Instructor	Mon, Wed 3:00 – 4:00 PM Old Chem 216 anya.katsevich@duke.edu
Gwen Jacobson	Head TA	Fri 1:00 – 2:00 PM, 5:00 – 6:00 PM Old Chem 203B
Federico Lazzarini	TA	Tue, Thu 5:00 – 6:00 PM Old Chem 203B

Course Materials

Textbooks

You are not required to purchase a textbook for this class. However, if you wish to follow along with one, these are all good options:

- [Ross] *A First Course in Probability* by Sheldon Ross
- [HTZ] *Probability and Statistical Inference* by Robert Hogg, Elliot Tanis, and Dale Zimmerman
- [DS] *Probability and Statistics* by Morris DeGroot and Mark Schervish
- [Wass] *All of Statistics* by Larry Wasserman

Technology

Lecture will largely be a low-tech affair: pencil and paper should be sufficient most of the time. You should always plan to bring a laptop or tablet device to lab. In general, you will need access to a device with internet so that you can use:

- This course website
- R/RStudio via the Duke Container Manager
- Canvas (which provides access to Gradescope and Ed Discussion)
- Zoom (e.g. for remote office hours)

If access to technology becomes a concern for you during the semester, contact the instructor immediately to discuss options.

Assignments and Grading

Your final course grade will be calculated as follows:

Category	Percentage
Labs	10%
Problem Sets	21%
Midterm Exam 1	23%
Midterm Exam 2	23%
Final Exam	23%

Final letter grades are determined by:

Grade	Range	Grade	Range	Grade	Range
A+	≥ 97	B+	87 – 89.99	C+	77 – 79.99
A	93 – 96.99	B	83 – 86.99	C	73 – 76.99
A-	90 – 92.99	B-	80 – 82.99	C-	70 – 72.99
D+	67 – 69.99	D	63 – 66.99	D-	60 – 62.99
F	< 60				

These thresholds will not change and will be applied exactly. Final grades will not be curved, and a 92.99 will not be rounded up to an A.

Labs (10%)

In lab every Thursday, you will complete a guided activity that introduces you to special topics, extensions, or applications of the latest course material, including implementation in R. Labs are due by 11:59 PM ET the same day they are held.

Each lab is graded:

- 1.0 – a complete, good faith attempt at every part of every problem
- 0.5 – partially complete
- 0.0 – no submission, or mostly incomplete

Grace

Your two lowest lab scores will be dropped at the end of the semester.

Problem Sets (21%)

Problem sets are the heart of the course. They mostly consist of pencil-and-paper math problems (sometimes with coding). You may compose solutions however you like so long as you ultimately upload a single .pdf file to Gradescope.

Grace

Your lowest problem set score will be dropped at the end of the semester.

Exams (23% each)

There will be three exams. Dates, times, and locations are firm:

- **Midterm 1:** Wednesday, February 18 during lecture
- **Midterm 2:** Wednesday, April 1 during lecture
- **Final:** Monday, April 27, 9:00 AM – 12:00 PM, Old Chem 116

These are old-school, pencil-and-paper, in-class exams. The only allowed resource is both sides of one 8.5" × 11" sheet of notes, handwritten by you alone (directly onto paper, not on a tablet).

If you seek testing accommodations, make sure the Student Disability Access Office sends the instructor a letter, and make your appointments in the Testing Center as soon as possible.

Grace

If you do better on the final exam than on one of the midterms, your lowest midterm score will be replaced with your final exam score.

No make-up exams

If you miss a midterm for any reason, there will be no make-up — the missed midterm score will simply be replaced by the final exam score. If you miss the final exam, you receive a zero unless you have a Dean's Excuse.

Policies

Collaboration

You are *enthusiastically encouraged* to work together and help one another on labs and problem sets. What is asked is that you *acknowledge* your collaborators. So, at the end of each problem in your write-up, leave a note such as "Ursula, Ignatius and I worked together on this problem." You are not judged based on your acknowledgements, and there is no penalty for getting "too much" help from others.

Having said that, you should not be handing your solutions to others for them to brainlessly copy. This is plagiarism, and all involved will earn a zero and be referred to the conduct office. The write-up you submit must be your own work.

Use of Outside Resources, Including AI

There are at least two reasons you might seek outside resources:

- **Extra practice or alternative instruction:** Go crazy. The internet is saturated with good (and horrible) resources for learning this material.
- **Doing the problems for you:** If you find a solution online, or ask a language model to generate one, and you copy it and submit it as your own work, that is plagiarism. If detected, you will earn a zero for that part of your write-up.

Furthermore, 69% of your final course grade is determined by your performance on old-school, no-tech exams. Outsourcing your thinking to an AI will probably end in disaster.

Communication

For content-related questions in writing, please use the course discussion forum **Ed Discussion** (not e-mail), so all members of the teaching team and all students can benefit. For personal matters (illness, accommodations, etc.), e-mail the instructor directly at anya.katsevich@duke.edu.

> You can ask questions anonymously on Ed. The teaching team will still know your identity, but your peers will not.

Late Work and Extensions

No late work will be accepted unless you request an extension *in advance* by e-mailing the instructor directly. All reasonable requests will be entertained, but extensions will not be long.

Regrade Requests

If you believe part of a graded assignment was marked incorrectly, you may submit a regrade request in Gradescope. Note the following:

- You have one week after receiving a grade to submit a regrade request;
- Submit separate regrade requests for each question you wish to dispute;
- Requests will be considered if there was an error in the grade calculation or if a correct answer was mistakenly marked as incorrect;
- Requests to dispute the number of points deducted for an incorrect response will not be considered;
- **No grades will be changed after the final exam on Monday, April 27.**

Attendance

Attendance is not strictly required for any class meetings. The responsibility lies with the teaching team to make class sufficiently engaging and informative that you choose to attend. That said, success in this class and regular attendance are probably highly positively correlated. While lab attendance is not required, showing up, completing the lab in one sitting, and submitting by midnight is probably the path of least resistance to full lab credit.

Accommodations

If you need accommodations, register with the Student Disability Access Office (SDAO) and provide documentation. SDAO will work with you to determine appropriate accommodations. Note that accommodations are not retroactive and cannot be provided until a Faculty Accommodation Letter has been given to the instructor. Contact SDAO: sdao@duke.edu or access.duke.edu.

Duke Community Standard

Duke University is a community dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflect upon and uphold these principles in all academic and non-academic endeavors, and to protect and promote a culture of integrity.

In accepting admission, students indicate their willingness to subscribe to and be governed by the rules and regulations of the university, which flow from the Duke Community Standard (DCS). It is the responsibility of all students to understand and follow all Duke policies, including the academic integrity policy.

In STA 240 specifically:

- If a conduct violation results in a zero on a lab or problem set, that zero will not be dropped;
- If a conduct violation results in a zero on a midterm, that zero will not be replaced with your final exam score;
- If students are found sharing and copying assignment solutions, all students involved will be penalized equally.

University Resources

Course Costs

If you are having difficulty with costs associated with this course (obtaining a laptop, mostly), here are some resources:

- **Karsh Office of Undergraduate Support:** Offers loans and resources for connecting students with campus programs.
- **DukeLIFE:** The Course Material Assistance program offers assistance for eligible students, including the LIFE Loaner Laptop Program.
- **Duke Link:** Has a small supply of laptops that can be rented out for five days at a time.

Tech Support

Contact the Duke OIT Service Desk at oit.duke.edu/help.

Academic Support

The Academic Resource Center (ARC) offers services to support students academically, including time management, academic skills and strategies, and course-specific tutoring. ARC services are free to all Duke undergraduate students. Contact: (919) 684-5917, theARC@duke.edu, or arc.duke.edu.

Accessibility

If any portion of the course is not accessible to you due to challenges with technology or the course format, please let the instructor know so appropriate accommodations can be made. The Student Disability Access Office (SDAO) is available to ensure students can engage with their courses and related assignments.

Mental Health and Well-Being

Duke is committed to holistic student wellbeing. If you find you need support, resources include:

- **DukeReach:** Comprehensive outreach services for students managing challenges related to mental health, physical health, social adjustment, and other stressors. Contact: dukereach@duke.edu.
- **Counseling and Psychological Services (CAPS):** Counseling services including virtual appointments. Walk in or call (919) 660-1000. Hours: Mon–Fri 9:00 AM – 4:00 PM. After-hours: (919) 660-1000 Option 2.
- **TimelyCare:** Free 24/7 mental health support (TalkNow and scheduled counseling) for Duke students.
- **Duke Student Health:** Wide range of healthcare services. Call (919) 681-9355. Hours: Mon–Fri 8:00 AM – 4:30 PM (Thu 9:00 AM – 4:30 PM).